ADAGIO GROUP

Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Paper 2: Koopman Spectral Theory - My Response to Sara's Latest Comments and Ezequiel's Review

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 24, 2026 at 1:21 AM
To: "ISMAIL-SUTTON, SARA R." <sara.r.ismail-sutton@durham.ac.uk>, Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Sara and Ezequiel,

Below I first address Sara's comments on my response to her manuscript review, then below I address Ezequiel's review as I had failed to previously. Thank you both for your efforts.

On Sara's comments…

Thank you for the follow-up. Revision 5 (https://dx.doi.org/10.2139/ssrn.6733181) implements the four open points.

The 1/3 bound now carries the contraction assumption at every statement, including the two sentences that lacked it (the lead-in to Corollary 4.6 and the Discussion summary).

The augmentation passage names the constraints and the theorems: Bell locality, excluded by Bell's theorem in Fine's form (a joint distribution over all settings exists iff CHSH holds); noncontextuality, excluded by Kochen–Specker in dimension at least three. Bell, Fine, and Kochen–Specker are cited; the sentence that the paper does not establish non-augmentability stays.

Theorem 4.5(a) is now marked, in the paragraph before it, as a structural consequence of the subspace definitions and the Schmidt decomposition, with the dynamical content located in (b), Corollary 4.6, Proposition 4.12, and Theorem 7.4.

Table 1 in the related-work section gives the theorem-level comparison: for each of the three programs, what it presupposes, which results here have a counterpart in it, and which have none. If it misses a comparison you had in mind, tell me which.

On Step 1b: by (R1), P(s,s) is exactly the projection onto H⊗|0⟩, so I − P(t,s) has constant rank on the closed simplex; Steenrod triviality over the contractible base gives a continuous frame of its range; rotating that frame by $G(s)^{-1}$, where $f_\alpha(s,s) = \Sigma_\beta G_{\beta\alpha}(s)\, e_\beta$ with G continuous in s, makes it equal the reference frame on the diagonal, so W(s,s) = I.

Thank you for withdrawing the (R3′) point.

On Ezequiel's review…

Thank you for the evaluation. Your report examined the September 3 deposit; revision 6, enclosed, post-date it. Revision 5 (https://dx.doi.org/10.2139/ssrn.6733181) implements every item except the title, which stays, and Question 7, which I answer below rather than in the paper.

Finding 4, Question 2. Confirmed. For the family Γ(T,T/2) = Γ(T/2,0) = I, Γ(T,0) = I − (D/2)(I − X), every divisible column-stochastic comparison has sup δ ≥ m(D) = (3 − √(9 − 4D))/2, and the semigroup with $e^{-\kappa T/2} = 1 - m(D)$ attains it; $m(D)/(D/3) = 1 + m(D)^2/D$, which reproduces your five ratios and tends to 1. The lower bound holds for all column-stochastic comparisons, not only the symmetric reduction: a column-stochastic 2×2 matrix has $A - I = v\, w^{\top}$ with ⟨w, v⟩ ≤ 0, so the product term of AB opposes the deficit for every admissible pair. The result is Proposition 4.7, with the family extended to continuous time so that the statement holds on the whole simplex, and the former Remark 4.7 is closed. The bound is strict for every D > 0 and is approached only at deterministic intermediate kernels; the paper says so.

Finding 1, Question 1. The identity $C = A' + A^2$ for $\partial_t U = AU$ is stated with Dyson and Blanes–Casas–Oteo–Ros cited, followed by a sentence saying what the section adds: the residual as a named invariant, the factor (t − r)(r − s), the singular-direction reading, and the place of β in the global theory.

Finding 5, Question 4. Canturk and Doukas are cited in the related-work section and at the sixth contribution, and Section 5 no longer claims the basic lift. What remains is the diagonal-retention channel with Φ(s,s) = id, the jointly continuous completion with W(s,s) = I on a fixed environment of dimension $n^2 - n + 1$, the formal generator under (R3′), the grid cocycle, and the incorporation into the deficit framework.

Finding 2, Question 5. The remark now reads "a formal Hamiltonian generator can be constructed for a selected continuous dilation" and states the gauge dependence: under $W \mapsto (I \otimes U(t))W$ the channel is unchanged and $\hat{H}$ transforms by conjugation plus $i\dot{U}U^*$, so no feature of $\hat{H}$ is invariant; the invariants are the channel family, the populations, and the deficit.

Proportion and priority. Proposition 3.6 and Corollary 4.6 carry the scope-fixing label already used for Proposition 2.6, Theorem 5.4 has a lead-in separating the imported machinery (Stinespring, Kretschmann–Schlingemann–Werner, Kato, Steenrod) from the diagonal-retention indexing, and the sewing and almost-flow literature is acknowledged in a new related-work paragraph.
Question 3. The nearest notions I know are the Sz.-Nagy–Foiaş defect spaces of a single contraction, the sewing defect $\mu_{ts} - \mu_{tu} \circ \mu_{us}$ of almost-flows, and Tian's defect kernel of a compressed semigroup (June 2026). None spans the ranges of all composition deficits over all triples; that global locus, with vanishing on $V_{in}^{\perp}$ and the approximation obstruction of Theorem 4.5(b), is the claim I would defend. The pointwise range and kernel facts are linear algebra.

Question 6. Theorem 5.4(iii) is a collision model with a fresh environment factor per grid increment, and Table 1 lists which results have process-tensor counterparts. The distinctions are classical input, a fixed environment per pair, and joint continuity in (t,s).

Question 7. Estimate $\hat{\Gamma}$ on a time grid from paired observations, form $\hat{\Delta}$ on triples, and test $\Delta \equiv 0$ by parametric bootstrap under the fitted product family, with statistic max over triples of $\hat{D}$; estimate β from the slope of $\hat{\Delta}$ against $(t - r)(r - s)$ over near-diagonal triples rather than from second differences of $\hat{\Gamma}$. Conditioning: $\Delta \sim h^2\|\beta\|$ while $\hat{\Gamma}$ carries error of order $N^{-1/2}$, so relative precision ρ on β needs $N \gtrsim (\rho\, h^2\, \|\beta\|)^{-2}$ transitions per pair; the h^4 is intrinsic to a second-order quantity, and the regression removes only the cancellation between separately estimated coefficients. The paper keeps the estimator open.
Question 8. A measured deficit establishes non-divisibility of the chosen description and the absence of a Dyson-class generator for it, not intrinsic memory: any finite-memory process is a lumping of a Markov process on an augmented space, and a compressed semigroup has a nonzero defect. Remark 2.8 and Section 8 say so.

Question 9. New Section 6.8: the theorems carry no dimensional hypothesis, the two examples witness nonvacuity in closed form, and a product rule, $\Delta_{\Gamma\otimes\tilde{\Gamma}} = \Delta \otimes \tilde{\Gamma}(t,s) + \Gamma(t,r)\Gamma(r,s) \otimes \tilde{\Delta}$ and $\beta_{\Gamma\otimes\tilde{\Gamma}} = \beta \otimes I + I \otimes \tilde{\beta}$, supplies closed-form families in dimension 2^k. The product of two copies of the two-state family is a four-state family with a rank-three deficit, $V_{in} = V_{out}$ the mean-zero subspace, and leading direction $v \otimes v$.

Ben

Benjamin D. Summers
Managing Director
ADAGIO GROUP
MOBILE +1.713.775.1011 | **OFFICE** +1.305.433.1823 | **FAX** +1 239.932.5558
bsummers@adagiollc.com | adagio.capital